

TRADE AND TARIFFS

PATHWAYS TO ARTS & HUMANITIES SUMMER SCHOLARS PROGRAM

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Summer 2026

REVIEW: TRAGEDY OF THE COMMONS

Question

A town has a public lake where anyone can fish for free. Over time, the fish population declines sharply. Which combination of properties makes the fish a **common resource**?

- A. Rivalrous and excludable
- B. Non-rivalrous and non-excludable
- C. Rivalrous and non-excludable
- D. Non-rivalrous and excludable

REVIEW: TRAGEDY OF THE COMMONS — ANSWER

Answer

The correct answer is **C. Rivalrous and non-excludable**.

The fish are **rivalrous**: one person's catch reduces what is available for others.

The lake is **non-excludable**: anyone can fish without paying.

This combination creates the Tragedy of the Commons — overfishing depletes the resource.

CLAIM 1

Question

Thumbs up (agree), thumbs down (disagree), or sideways (not sure):

“International trade is generally good for a country’s economy.”

CLAIM 2

Question

Thumbs up (agree), thumbs down (disagree), or sideways (not sure):

“International trade is generally good for every sector of a country’s economy.”

CLAIM 3

Question

Thumbs up (agree), thumbs down (disagree), or sideways (not sure):

“Tariffs protect a country when international trade is bad for the economy.”

We'll revisit these at the end of class.

OUTLINE

1. Is Trade Good for a Country?

2. Does Everyone Benefit?

3. Are Tariffs the Answer?

4. Recap

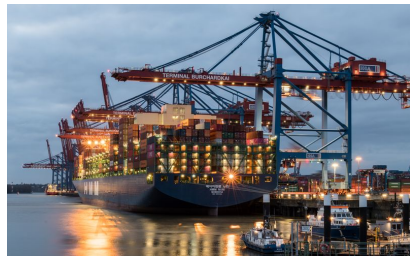
WHAT IS INTERNATIONAL TRADE?

International Trade

The exchange of goods and services across national borders.

Imports = goods coming IN to a country (*your phone, made in China, imported to the U.S.*)

Exports = goods going OUT of a country (*U.S. soybeans, shipped to China*)



WHY DO COUNTRIES TRADE?

Not every country can make everything well:

- Different climates, resources, and skills
- Some countries have more workers; others have more technology
- Making everything yourself is expensive and slow



A TALE OF TWO COUNTRIES

Question

Imagine a country that *is* good at everything. Should it trade at all?

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The setup: Each country has 100 workers.

United States

Good at producing both software AND coffee
Especially productive at software

Colombia

Less productive at both
But relatively better at coffee

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With 100 workers, each can produce:

	Software (units)	Coffee (tons)
U.S.	40	20
Colombia	10	20

THE KEY: OPPORTUNITY COST

Opportunity Cost (OC)

The **value** of the **next best alternative** when you make a decision.

With 100 workers, each can produce:

	Software (units)	Coffee (tons)	OC of Coffee (software units)	Benefit/OC (units/ton)
U.S.	40	20	40	$\frac{1}{2}$
Colombia	10	20	10	2

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The Benefit / Opportunity Cost ratio of producing Coffee is much higher for Colombia than it is for the US! We say that Colombia has a *comparative advantage* in coffee.

COMPARATIVE ADVANTAGE

Comparative Advantage

Being **relatively better** at producing something—having a higher benefit / opportunity cost ratio—even if you are not the most productive in absolute terms.

	Comp. Advantage	Why?
United States	Software	Benefit / OC of software = 2 unit / ton
Colombia	Coffee	Benefit / OC of coffee = 2 ton / unit

NO TRADE: EACH COUNTRY MAKES EVERYTHING

If each country splits its workers 50/50 between software and coffee, then the countries would produce:

	Software (units)	Coffee (tons)
United States	20	10
Colombia	5	10
World total	25	20

Not bad—but can they do better?

WITH TRADE: SPECIALIZE AND SWAP

Step 1: Specialize

Each country puts all workers on its comparative advantage:

	Software (units)	Coffee (tons)
U.S.	40	0
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Step 2: Trade

The U.S. gives Colombia 10 software units for 10 tons of coffee.

	Software (units)	Coffee (tons)
U.S.	30	10
Colombia	10	10

GAINS FROM TRADE

No Trade (50/50 split)

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With Trade (specialize + swap)

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Same coffee, but **15 more software units** for the world. The U.S. gains +10, Colombia gains +5. **Both countries are better off!**

*Even though the U.S. is more productive at **both** goods, trade still helps—each country specializes in what it does relatively best.*

CLAIM 1: IS TRADE GOOD FOR A COUNTRY?

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Answer

Yes—international trade generally grows the economic pie. Through specialization and comparative advantage, countries produce more together than they could alone.

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A BIGGER PIE, BUT WHO GETS THE SLICES?

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Full Specialization

	Software (units)	Coffee (tons)
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Assumption: Workers can be moved from one sector to another **without any cost**.

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Question

What is the problem with this assumption?

COSTS OF SPECIALIZATION

Question

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Answer

- Workers in shrinking sectors **lose their jobs** in the short run
- They need to be **retrained** for a new job, which takes time and money
- If the new job requires higher skills, they may also need a **college education**
- If the new job is in a different city, they need to be **highly mobile**, i.e. willing and able to relocate

DEINDUSTRIALIZATION IN AMERICA

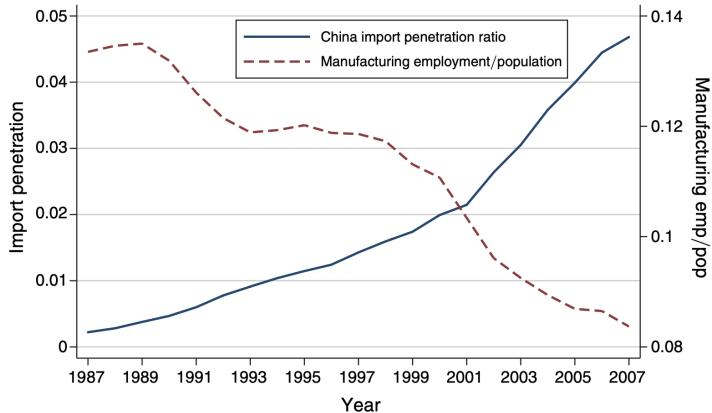


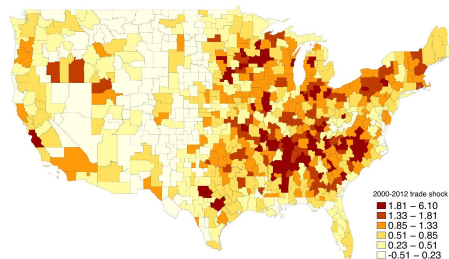
FIGURE 1. IMPORT PENETRATION RATIO FOR US IMPORTS FROM CHINA (*left scale*), AND SHARE OF US WORKING-AGE POPULATION EMPLOYED IN MANUFACTURING (*right scale*)

THE CHINA SHOCK

Economists studied what happened to U.S. communities most exposed to Chinese imports (Autor, Dorn & Hanson 2021 BPEA)

- China's entry to the WTO (2001) flooded the U.S. with cheap manufactured goods
- Regions most exposed saw persistent **job losses** in manufacturing, **lower income per capita**
- Net out-migration only for foreign-born workers and young native born

Figure 4: Chinese Import Penetration (2000-2012) in U.S. Commuting Zones



Source: Autor, Dorn & Hanson (2021), Figure 4

CLAIM 2: DOES EVERY SECTOR BENEFIT?

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CLAIM 2: DOES EVERY SECTOR BENEFIT?

Question

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Answer

No—not every sector benefits from trade. Trade creates winners and losers. Import-competing industries shrink, and workers in those sectors face persistent job losses. Transition to growing sectors is possible in theory, but difficult, slow, and painful in practice.

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WHO BEARS THE COST OF THE TARIFF?

Base case: A TV imported from China and sold by Best Buy.



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Option B — Tariff paid by the importer:

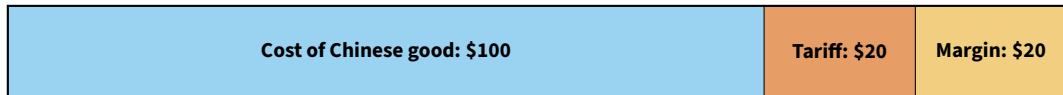


WHO BEARS THE COST OF THE TARIFF?

Base case: A TV imported from China and sold by Best Buy.



Option C — Tariff paid by the consumer:



PASS-THROUGH OF TARIFFS

Pass-Through

The share of a tariff increase that is passed on to consumers through higher prices.

- **Option A** (exporter pays): pass-through = 0% — the exporter absorbs the cost
- **Option B** (importer pays): pass-through = 0% — the importer absorbs the cost
- **Option C** (consumer pays): pass-through = 100% — the full tariff shows up in the price

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Answer

≈ **100%**

The Budget Lab at Yale, Jan 2026.

WHY DO COUNTRIES USE TARIFFS?

Economic

Protect nascent industries

Germany's Zollverein (1834) shielded young industries from British competition

Security

Protect strategic industries

U.S. CHIPS Act (2022): tariffs + subsidies to secure semiconductor production

Political

Protect political industries

EU agricultural tariffs protect French farmers despite cheaper imports abroad

Strategic

Bargaining leverage

Use tariff threats to pressure partners into lowering their tariffs (e.g. USMCA)

CASE STUDY: STEEL TARIFFS (2018)

In 2018, the U.S. government placed a **25% tariff on imported steel** and a **10% tariff on imported aluminum**.

- **What:** A tax on steel and aluminum coming into the U.S.
- **Who it applied to:** Nearly every country that sells steel to the U.S.
- **Why:** The government argued that the U.S. needs its own steel industry for **national security**

Question

What are the benefits and costs?

VIDEO: THE STEEL TARIFF DEBATE

Steel Tariff Debate



[Click to watch the video](#)

STEEL TARIFFS: BENEFITS AND COSTS

Arguments made in the video:

Benefits

- **Small businesses:** Small domestic steelmakers can compete better against cheap imports
- **Reciprocal trade:** China was “dumping” steel (selling below cost to destroy competitors), this tariff acts as a disincentive for steel and other goods
- **National security:** The U.S. needs to be able to make its own steel, this tariff promotes important industry

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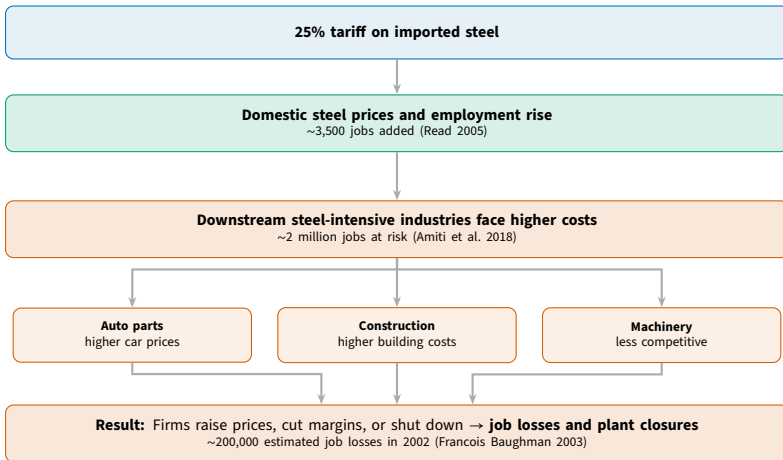
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Costs

- **Losses > Gains:** “All studies on the 2002 steel tariffs find that the costs of the tariffs outweighed their benefits in terms of GDP and employment”
- **Higher consumer prices:** Consumers end up paying more for industries relying on steel inputs
- **Wrong target:** Chinese steel is only **2%** of U.S. steel imports, so if anything, these tariffs are ineffective

STEEL TARIFFS: ZOOM IN ON EMPLOYMENT



STEEL TARIFFS: WHAT ABOUT TRADE WARS?

The video discussion did **not** account for trade wars!

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WORLD • ASIA • CHINA

China Retaliates Against Trump Tariffs With Duties on American Meat and Fruit

Penalties range from 25% on American pork and eight other kinds of goods to 15% on fruit and 120 types of commodities

By [Charles Hutzler](#)

Updated April 1, 2018 10:47 pm ET



Aa



828



Gift unlocked article

When the U.S. raised tariffs, trading partners hit back.

American exporters—who had *nothing to do with steel*—paid the price.

CLAIM 3: ARE TARIFFS A GOOD FIX?

Question

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Answer

It's complicated. Tariffs can protect specific *sectors*, but they raise costs for consumers and downstream firms. The costs often outweigh the benefits. Retaliation hurts unrelated industries.

There is no free lunch—every policy involves tradeoffs.

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LET'S REVISIT OUR THREE CLAIMS

1. “International trade is generally good for a country’s economy.”
→ **Yes**—comparative advantage grows the pie.
2. “International trade is generally good for every sector of a country’s economy.”
→ **No**—creates winners and losers. Transition is hard.
3. “Tariffs protect a country when international trade is bad for the economy.”
→ **It’s complicated**—protects some, taxes others. Costs often exceed benefits.

Has your thinking changed since this morning?